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324812(24)

B. E. (Eighth Semester) Examination, April-May 2016

(Old Scheme)

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(Elect., EEE Branch)

UTILIZATION of ELECTRICAL ENERGY

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *All questions are compulsory. Part (a) is compulsory and solve any* ***two*** *out of (b), (c) and (d).*

Unit-I

1. (a) Discuss the advantages of electric drives. 2

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(b) What is load equalisation? Where do you find its applications? Derive the expression for instantaneous

motor torque, MOI of the flywheel, and the motor slip. State any assumptions made. 7

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(c) Describe four quadrant operation of a motor driving a hoist load with help of a labelled diagram showing direction of motor torque, load torque and motor speed, as applicable in each quadrant. 7

(d) A motor equipped with a flywheel is to supply a load torque of 1000 N-m for 10 sec. followed by a light load period of 200 N-m long enough for the flywheel to regain its steady state speed. It is desired to limit the motor torque to 700 N-m. What should be the moment of inertia of flywheel? Motor, has an inertia of 10 kg-m^2. Its no load speed is 500 rpm and the slip at a torque of 500 N-m is 5%. Assume speed-torque characteristic of motor to be a straight line in the region of interest. 7

Unit-II

(a) What is Duty Cycle? Explain. 2

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(b) How does a phase-locked loop speed control scheme operate? Where do you use it? 7

(c) What are heating and cooling time constants? What are the factor on which it depends and how? 7

(d) A motor has a heating time constant of 60 min and cooling time constant of 90 min. When run continuously on full load of 20 kW, the final temperature rise is 40°C. 7

(i) What load motor can deliver for 10 min. If this is followed by a shutdown period long enough for it to cool?

(ii) If it is on an intermittent load of 10 min followed by 10 min shutdown, what is the maximum value of load it can supply during the on load period?

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Unit-III

3. (a) Give some application of cyclo convertor. 2

(b) How the speed and power factor of a wound rotor induction motor are controlled by injecting a voltage in the rotor circuit? What should be the relation between the frequency of the injected voltage and the frequency of the rotor induced voltage. 7

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(c) Explain the operation of a self controlled synchronous

motor drive employing a load commutated thyristor inverter. 7

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(d) Describe static Kramer drive scheme. 7

Unit-IV

4. (a) Why 3 ϕ induction motor are not used in electric traction? 2

(b) Discuss 25 kV ac traction drive employing thyristor converter controlled d.c. motors. Why a converter with sequence control is employed? 7

(c) Explain the various types of resistance and arc welding. How the mechanical pressure, amount of current and its duration is controlled in case of resistance welding? 7

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(d) An electric train has an average speed of 42 kmph on a level track between stops 1400 m apart. It is accelerated at 1·7 kmphps and is braked at 3.3 km phps. Draw the speed-time curve for the run. 7

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Unit-V

5. (a) Define the following terms : 2

(i) Luminous intensity

(ii) Illumination

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(b) Explain how the use of variable speed drive will improve energy conservation. 7

(c) Derive expression for the illumination on a surface (i) when it is normal and (ii) when it is inclined to the axis of a beam of incident light. 7

(d) Describe in brief the requirements of good lighting. Enumerate the factors to be considered while designing an indoor lighting scheme. 7

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